

Mixed Number Conversion Trail

Unit 2: Fraction Detective Agency • Lesson 2-4 • EduWonderLab

UNIT 2 • LESSON 2-4 • TEACHER NOTES

Standard 6.NS.1

FORMAT Self-checking trail / maze

TIME 25–30 min

GROUPING Pairs or solo

TOPIC Divide Mixed Numbers

STANDARD 6.NS.1

PREP LEVEL Light — print trail cards and hint cards

Learning goal *I can divide mixed numbers by first writing them as improper fractions.*

Teacher setup

- Print one trail per pair. Each card converts, rewrites as multiplication, multiplies, and simplifies.
- A wrong turn sends students to a hint station before trying again.
- Answer choices appear only at the final station.

Materials

Trail cards, Improper-fraction conversion box, Hint cards, Pencil.

Differentiation & language support

- Conversion box shows how to turn a mixed number into an improper fraction (SPED).
- Sentence stem: “I rewrote ___ as ___ before I divided.”
- ESOL: keep one fully worked example visible on the trail.

Key vocabulary (English / Español):

Term	Español	What it means
mixed number	número mixto	a whole number and a fraction together
improper fraction	fracción impropia	a fraction with a numerator larger than its denominator

Quick teacher check: *Check the conversion step on Card 4 ($4\frac{1}{2} \div 1\frac{1}{2} \rightarrow \frac{9}{2} \div \frac{3}{2}$).*

Mixed Number Conversion Trail

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Name: _____

Date: _____

Class: _____

Your goal: *I can divide mixed numbers by first writing them as improper fractions.*

What you need

Trail cards, Conversion box.

How to play

1. Start at Card 1. Convert each mixed number to an improper fraction.
2. Rewrite the division as multiplication by the reciprocal.
3. Multiply and simplify. Write the answer as a mixed number when possible.
4. Check your answer before moving to the next card.

Trail Cards

Convert each mixed number to an improper fraction first, then divide with the reciprocal.

1. $2\frac{1}{2} \div \frac{1}{4}$	2. $3\frac{1}{3} \div \frac{2}{3}$
3. $1\frac{3}{4} \div \frac{1}{2}$	4. $4\frac{1}{2} \div 1\frac{1}{2}$
5. $2\frac{2}{3} \div 1\frac{1}{3}$	6. $3\frac{3}{4} \div 1\frac{1}{4}$
7. $5 \div 1\frac{1}{4}$	8. $2\frac{1}{4} \div \frac{3}{8}$
9. $6\frac{2}{3} \div 2$	10. $1\frac{1}{2} \div 2\frac{1}{4}$

Record your work (show the improper-fraction step):

Explain your thinking

Explain why you must convert mixed numbers to improper fractions before dividing.

Sentence frame: *I changed ___ to the improper fraction ___ so I could multiply by the reciprocal.*

★ **Challenge:** *Card 10 gives an answer less than 1. Explain how the dividend and divisor caused that.*

ANSWER KEY

Teacher copy — please do not distribute to students.

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Answers

1. $2\frac{1}{2} \div \frac{1}{4}$ Answer: 10	2. $3\frac{1}{3} \div \frac{2}{3}$ Answer: 5
3. $1\frac{3}{4} \div \frac{1}{2}$ Answer: $3\frac{1}{2}$	4. $4\frac{1}{2} \div 1\frac{1}{2}$ Answer: 3
5. $2\frac{2}{3} \div 1\frac{1}{3}$ Answer: 2	6. $3\frac{3}{4} \div 1\frac{1}{4}$ Answer: 3
7. $5 \div 1\frac{1}{4}$ Answer: 4	8. $2\frac{1}{4} \div \frac{3}{8}$ Answer: 6
9. $6\frac{2}{3} \div 2$ Answer: $3\frac{1}{3}$	10. $1\frac{1}{2} \div 2\frac{1}{4}$ Answer: $\frac{2}{3}$

Teaching notes & look-fors

- Card 10: $1\frac{1}{2} \div 2\frac{1}{4} = \frac{3}{2} \div \frac{9}{4} = \frac{2}{3}$ (less than 1 because dividend < divisor).
- Both numbers must be improper fractions before flipping.